## PROJECT SYNOPSIS REPORT

# CAMPUSPULSE

*"GP Patna-7 Smart Faculty Radar & Geofenced Attendance Terminal"*

A Diploma Project Proposal Submitted in Partial Fulfillment of Requirements for the Diploma in Computer Science & Engineering


**SUBMITTED BY:**

**1. Sarvjit Kumar Ray**
Roll No: 1181824020
*Role: Team Leader & Core Security Specialist*

**2. Adarsh Raj**
Roll No: 1181824039
*Role: Frontend & UI/UX Developer*

**3. Anshu Kumar**
Roll No: 1181824040
*Role: Quality Assurance & Testing Lead*

**SUBMITTED TO (PROJECT GUIDE):**

**Project Supervisor:**
Prof.Arti Sinha.
Government Polytechnic, Patna-7

**Head of Department (HOD):**
Department of Computer Science & Engg.
Government Polytechnic, Patna-7



**Academic Session: 2026 – 2027**
**State Board of Technical Education (SBTE), Bihar**

## 1. INTRODUCTION

In modern educational institutions such as Government Polytechnic, Patna-7, efficient time utilization and verified instructional presence are critical for academic excellence. With numerous academic departments, laboratory complexes, and administrative blocks spanning the campus, the lack of real-time coordination tools causes significant daily friction for both faculty members and students.

**CampusPulse** is an automated, web-based campus utility platform engineered to solve two foundational on-campus challenges: unscheduled, unproductive searches for faculty members and proxy-prone, manual student attendance recording. By combining client-side real-time state broadcasting with browser-level high-precision GPS geofencing, CampusPulse delivers a transparent, lightweight digital terminal designed specifically around the daily workflow of GP Patna-7.

## 2. PROBLEM STATEMENT

The conventional administrative and classroom management systems currently employed in polytechnic colleges suffer from significant operational bottlenecks:

- **Unproductive Cabin Visits:** Students routinely walk across multi-story department buildings to consult professors, only to find cabins locked because the faculty member is engaged in classroom lectures, departmental meetings, or official off-campus duty. This leads to substantial loss of productive academic hours.
- **Proxy Attendance Vulnerabilities:** Paper roll-call sheets, manual registers, and static digital solutions (such as projection of static QR codes or shared Google Form links) are easily manipulated. Absent students routinely obtain links or photos from peers over messaging applications and mark attendance remotely from hostels, canteens, or off-campus locations.
- **Lack of Real-Time Administrative Visibility:** College administration lacks an instantaneous, unified monitoring console to observe current lecture conduct, faculty availability, and live student turnouts across departments simultaneously.

## 3. OBJECTIVES OF THE PROJECT

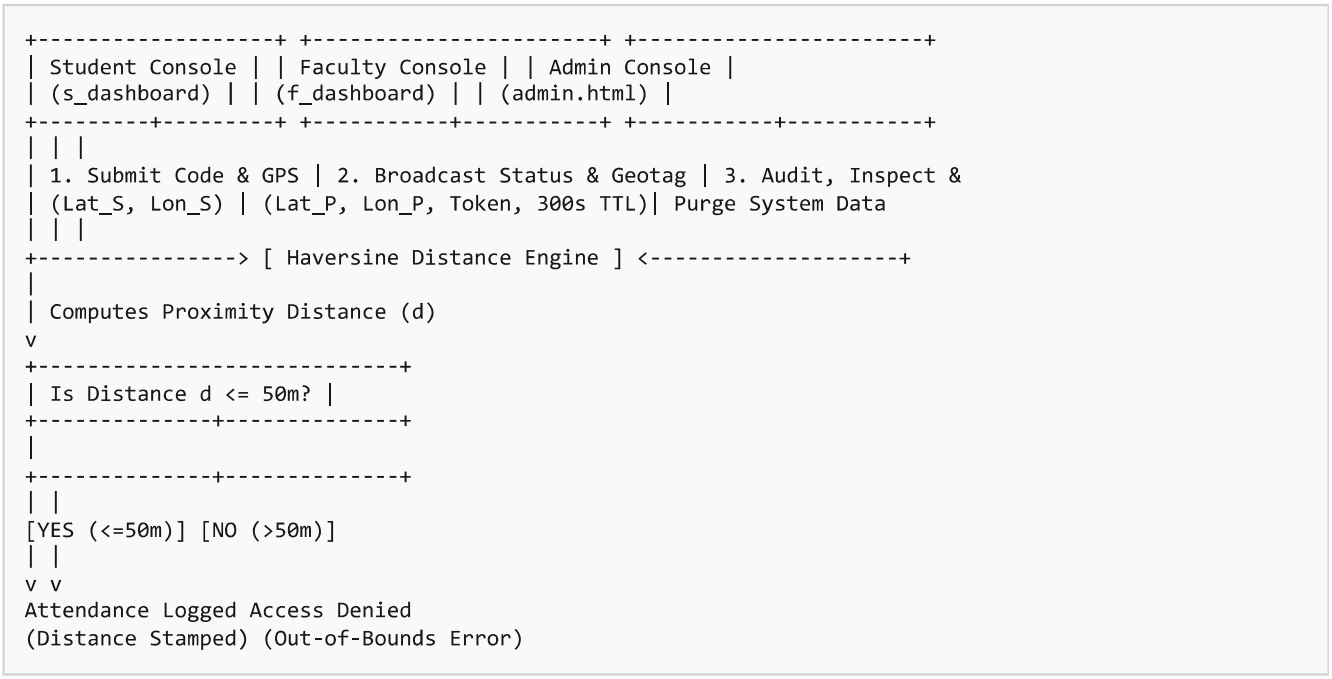
The core developmental and operational objectives of CampusPulse include:

1. **Real-Time Availability Radar:** Provide students with instant, color-coded visibility into each professor's live work state (*Free in Cabin, In Class, In Meeting, On Leave*) along with specific cabin designations.
2. **Enforced Geofenced Perimeter:** Implement strict geographic proximity boundary checking (limited to a precise 50-meter radius around the professor) to eliminate remote proxy submissions.
3. **Ephemeral Security Tokens:** Construct dynamic 6-digit session codes (e.g., `GPP7-XXXXXX`) featuring an automatic 5-minute (300 seconds) time-to-live (TTL) lifecycle, after which tokens are permanently invalidated.
4. **Atomic Anti-Duplicate Submissions:** Enforce composite validation keys combining student enrollment numbers and session token hashes to guarantee that one student cannot submit multiple records for a single lecture.
5. **Unified Administrative Management:** Provide institutional administrators with comprehensive control to audit registered students, manage faculty directories, and inspect cryptographically stamped attendance logs.

## 4. SYSTEM ARCHITECTURE & METHODOLOGY

CampusPulse operates as a modular, three-tier decentralized web terminal consisting of dedicated views for Administrators, Faculty Members, and Students:

### 4.1 Operational Architecture Workflow




### 4.2 Mathematical Model: The Haversine Formula

To compute the physical ground distance between the professor's device and the student's smartphone on a spherical Earth surface, the system executes the **Haversine Formula** within client-side JavaScript runtime:

$$\Delta\varphi = (lat2 - lat1) \cdot (\pi / 180)$$
$$\Delta\lambda = (lon2 - lon1) \cdot (\pi / 180)$$
$$a = \sin^2(\Delta\varphi / 2) + \cos(lat1 \cdot \pi/180) \cdot \cos(lat2 \cdot \pi/180) \cdot \sin^2(\Delta\lambda / 2)$$
$$c = 2 \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)})$$
$$d = R \cdot c \quad (\text{where } R = 6{,}371{,}000 \text{ meters})$$

If the computed scalar distance $d$ is less than or equal to **50 meters**, and the token verification matches within the active 300-second window, the attendance entry is accepted, appending enrollment data and verified accuracy metrics into the storage registry.

## 5. HARDWARE & SOFTWARE REQUIREMENTS

| Category | Component / Technology | Specification / Purpose |
|---|---|---|
| **Frontend UI/UX** | HTML5, CSS3, Tailwind CSS | Utility-first responsive framework for cross-device support |
| **Client Logic** | JavaScript (ES6+) | Haversine arithmetic, session timeouts, DOM manipulation |
| **Location Services** | W3C Geolocation API | High-accuracy GPS coordinate extraction (lat/long) |
| **Storage Architecture** | Client-Side Storage Engine | JSON-serialized relational schemas for prototype testing |
| **Client Hardware** | Mobile Smartphone or PC | Active GPS module, Chromium/WebKit/Gecko web browser |

## 6. IMPLEMENTATION SCHEDULE (GANTT CHART)

The project development lifecycle spans 8 discrete calendar weeks across the academic semester:

| Work Milestone / Phase | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 |
|---|---|---|---|---|---|---|---|---|
| Requirement Gathering, Campus Survey & Feasibility | ▮ | | | | | | | |
| UI/UX Layouts & Responsive Interface Design | | ▮ | ▮ | | | | | |
| Faculty Real-Time Availability Radar Module | | | ▮ | ▮ | | | | |
| W3C GPS Geolocation & Haversine Distance Engine | | | | ▮ | ▮ | | | |
| Dynamic Ephemeral Token Lifespan & Anti-Duplicate Lock | | | | | ▮ | ▮ | | |
| Master Admin Console, Data Auditing & Registry Binding | | | | | | ▮ | ▮ | |
| Field Testing at GP Patna-7 Campus & Final Synopsis Report | | | | | | | ▮ | ▮ |

## 7. TEAM MEMBERS & RESPONSIBILITY DISTRIBUTION

| Student Name | Enrollment No. | Assigned Project Role | Primary Deliverables |
|---|---|---|---|
| **Sarvjit Kumar Ray** | 1181824020 | Team Lead & Security Architect | Haversine coordinate mathematical logic, ephemeral token destruction (300s TTL), and composite anti-proxy locking mechanisms. |
| **Adarsh Raj** | 1181824039 | Frontend & UI/UX Developer | Tailwind CSS styling, responsive multi-role layout structuring, search/filtering tables, and visual status badging. |
| **Anshu Kumar** | 1181824040 | QA & Testing Lead | On-campus GPS boundary field verification at GP Patna-7, storage schema testing, edge-case analysis, and project documentation. |

## 8. EXPECTED OUTCOMES & SCOPE

Upon full deployment, **CampusPulse** will eradicate proxy attendance across classrooms, eliminate unnecessary physical visits to staff rooms, and provide complete operational visibility for the administration of GP Patna-7. Future development will include transitioning to a cloud-based Node.js/PostgreSQL backend and integrating campus Wi-Fi BSSID handshakes for multi-story indoor accuracy.